

Hyper tuning Using Keras

import keras_tuner as kt

```
def build_model(hp):
```

```
    model = Sequential()
```

```
    model.add(Input(shape=(8,)))
```

```
    num_layers = hp.Int("num_layers",  
                        min_value=1,  
                        max_value=5)
```

```
    units = hp.Int("units", min_value=8,
```

```
                  max_value=128, step=8)
```

```
    activation = hp.Choice("activation",
```

```
                           values=["relu", "tanh",  
                                   "sigmoid"])
```

```
    for i in range(num_layers):
```

```
        model.add(Dense(units, activation=activation))
```

```
        model.add(Dropout(dropout_rate))
```

```
    model.add(Dense(1, activation="sigmoid"))
```

```
    optimizer = hp.Choice("optimizer",
```

```
                          values=["adam", "rmsprop", "sgd"])
```

```
    loss = "binary_crossentropy"
```

```
    metrics = ["accuracy"]
```

```
    model.compile(optimizer=optimizer, loss=loss,  
                  metrics=metrics)
```

```
    return model
```